

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

## ***B.Tech. Degree V Semester Special Supplementary Examination September 2022***

### **ME 19-205-0503 MACHINING SCIENCE AND MACHINE TOOLS (2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

**Course Outcomes**

On successful completion of the course, the students will be able to:

- CO1: To study the geometry of single point and multipoint cutting tools and mechanics of chip formation.  
 CO2: To get the concept of mechanics of metal cutting.  
 CO3: To understand the principles of working, specifications, different types and different operations performed in lathe, drilling and shaping machines.  
 CO4: To learn the principle of working, specifications, different types and different operations performed in grinding and milling machines.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

**PART A**(Answer **ALL** questions)

|    |     | (8 × 3 = 24)                                                                                    | Marks | BL | CO | PO |
|----|-----|-------------------------------------------------------------------------------------------------|-------|----|----|----|
| I. | (a) | What are the conditions favourable for built up edge formation?                                 | 3     | L2 | 1  | 1  |
|    | (b) | Explain the relationship of tool temperature and tool life.                                     | 3     | L2 | 2  | 1  |
|    | (c) | Briefly explain the classification of the elements of mechanical transmission in machine tools. | 3     | L1 | 3  | 1  |
|    | (d) | What are the special mechanisms employed in machine tool feed boxes?                            | 3     | L2 | 3  | 1  |
|    | (e) | How is a lathe specified? Explain with a neat sketch.                                           | 3     | L1 | 3  | 1  |
|    | (f) | How do the capstan and turret lathes improve productivity?                                      | 3     | L2 | 3  | 1  |
|    | (g) | What are the limitations of centreless grinding process?                                        | 3     | L2 | 4  | 1  |
|    | (h) | Draw a neat sketch of a drill jig and mark the important parts.                                 | 3     | L1 | 4  | 1  |

**PART B**

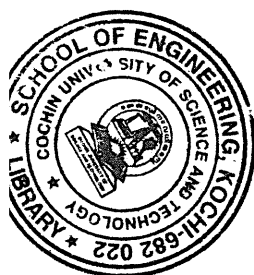
(4 × 12 = 48)

|     |     |                                                                                                              |   |    |   |   |
|-----|-----|--------------------------------------------------------------------------------------------------------------|---|----|---|---|
| II. | (a) | What are the conditions that would allow a continuous chip to be formed in metal cutting?                    | 3 | L2 | 1 | 1 |
|     | (b) | Derive the expression for shear angle in orthogonal cutting in terms of rake angle and chip thickness ratio. | 4 | L4 | 2 | 2 |
|     | (c) | Explain the geometry of a single point turning tool with the help of neat sketches.                          | 5 | L1 | 1 | 1 |

**OR**

|      |     |                                                                                                                                                                                                                                                                                                                                                      |   |    |   |   |
|------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| III. | (a) | Briefly explain the desirable characteristics of a cutting tool material.                                                                                                                                                                                                                                                                            | 3 | L2 | 1 | 1 |
|      | (b) | What are the various forms of wear found in cutting tools? Explain with the help of sketches.                                                                                                                                                                                                                                                        | 4 | L2 | 2 | 1 |
|      | (c) | In an orthogonal cutting of steel component with carbide tool the following data was obtained. Rake angle = 10°, chip width = 6 mm, uncut chip thickness = 0.1 mm, chip thickness ratio = 0.33, horizontal cutting force = 1290 N, vertical cutting force = 1650 N. Sketch the force diagram and calculate the mean shear stress on the shear plane. | 5 | L3 | 2 | 2 |

(P.T.O.)



BTS-V(SS)-09-22-1071

|           |                                                                                                                                                                                                                                                                       | Marks | BL | CO | PO |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| IV.       | (a) Explain the working and auxiliary motions in machine tools with suitable sketches.                                                                                                                                                                                | 6     | L2 | 3  | 1  |
|           | (b) Explain gear transmission used for transmitting rotory motion from one shaft to another in a machine tool.                                                                                                                                                        | 6     | L2 | 3  | 1  |
| <b>OR</b> |                                                                                                                                                                                                                                                                       |       |    |    |    |
| V.        | (a) What are the applications of hydraulic transmission in machine tools?                                                                                                                                                                                             | 6     | L2 | 3  | 1  |
|           | (b) What are the functions served by a lead screw in a machine tool?                                                                                                                                                                                                  | 6     | L2 | 3  | 1  |
| VI.       | (a) Compare the applications, advantages and disadvantages of a three jaw and four jaw chucks.                                                                                                                                                                        | 6     | L2 | 3  | 1  |
|           | (b) Sketch an example component suited for capston lathe with dimensions.                                                                                                                                                                                             | 6     | L3 | 3  | 2  |
| <b>OR</b> |                                                                                                                                                                                                                                                                       |       |    |    |    |
| VII.      | (a) Explain the quick return mechanism in a shaping machine.                                                                                                                                                                                                          | 6     | L2 | 3  | 1  |
|           | (b) Calculate the machining time required for shaping a 250 mm long and 200 mm wide flat surface on a shaper using a cutting speed of 40 m/min and a feed of 0.5 mm per stroke. The depth of cut is 4 mm. Calculate the material removal rate and the power required. | 6     | L3 | 3  | 2  |
| VIII.     | (a) Explain the tool geometry of a milling tool with the help of suitable sketches.                                                                                                                                                                                   | 6     | L2 | 4  | 1  |
|           | (b) What are the differences between compound indexing and differential indexing? Explain the relative merits.                                                                                                                                                        | 6     | L4 | 4  | 2  |
| <b>OR</b> |                                                                                                                                                                                                                                                                       |       |    |    |    |
| IX.       | (a) How is grinding different from other machining operations? Explain its applications in view of its capabilities.                                                                                                                                                  | 6     | L3 | 4  | 2  |
|           | (b) Explain an indexing jig with the help of suitable figures.                                                                                                                                                                                                        | 6     | L2 | 4  | 1  |

Bloom's Taxonomy Levels

L1 = 11.67%, L2 = 60.83%, L3 = 19.17%, L4 = 8.33%.

\*\*\*